

Incident Report: Multiple Failed Login Attempts

1. Incident Overview

Incident Title: Multiple Failed Login Attempts Detected

Detection ID: SOC-LAB-001

Severity: Medium

Tool Used: Splunk Enterprise

Data Source: Windows Security Event Logs

Index: soc_lab

Event ID: 4625

MITRE ATT&CK Mapping: T1110 - Brute Force

2. Alert Description

A Splunk alert was triggered after detecting multiple failed login attempts for the same user account within the monitored time range. Failed login activity can indicate password guessing, brute-force attempts, incorrect user credentials, or unauthorized access attempts.

This alert was created as part of a home-based SOC monitoring lab using Splunk Enterprise, Windows Security Logs, Sysmon, and Splunk Universal Forwarder.

3. Detection Logic

The detection rule searches for Windows Security Event ID 4625, which represents failed login attempts. The rule filters out invalid or anonymous account names and counts failed login attempts by account name, workstation name, and host. If the failed login count is equal to or greater than 3, the alert is triggered.

4. SPL Query Used

```
index=soc_lab sourcetype="WinEventLog:Security" EventCode=4625
| where Account_Name!="-" AND Account_Name!="ANONYMOUS LOGON"
| stats count as failed_count by Account_Name Workstation_Name host
| where failed_count >= 3
| sort - failed_count
```

5. Investigation Summary

The alert was investigated in Splunk Search & Reporting. The search results showed multiple failed login attempts related to the user account within the selected time range.

The following details were reviewed during investigation:

- Account name
- Hostname
- Workstation name
- Event ID
- Failed login count
- Event timestamp
- Triggered alert status

6. Observed Activity

Multiple failed login attempts were detected from the Windows endpoint. The failed login count met the detection threshold configured in the Splunk alert.

The alert was successfully triggered and appeared in the Splunk Triggered Alerts page.

7. Possible Cause

The activity may be caused by one of the following:

- User entered the wrong password multiple times
- Password guessing attempt
- Brute-force login attempt
- Cached credential issue
- Misconfigured service or application using old credentials

8. Impact Assessment

The activity does not confirm account compromise by itself. However, repeated failed login attempts can indicate suspicious authentication activity and should be reviewed by a SOC analyst.

Potential impact includes:

- Unauthorized access attempt
- Account lockout
- Password guessing activity
- Increased risk of account compromise if followed by successful login

9. Recommended Remediation

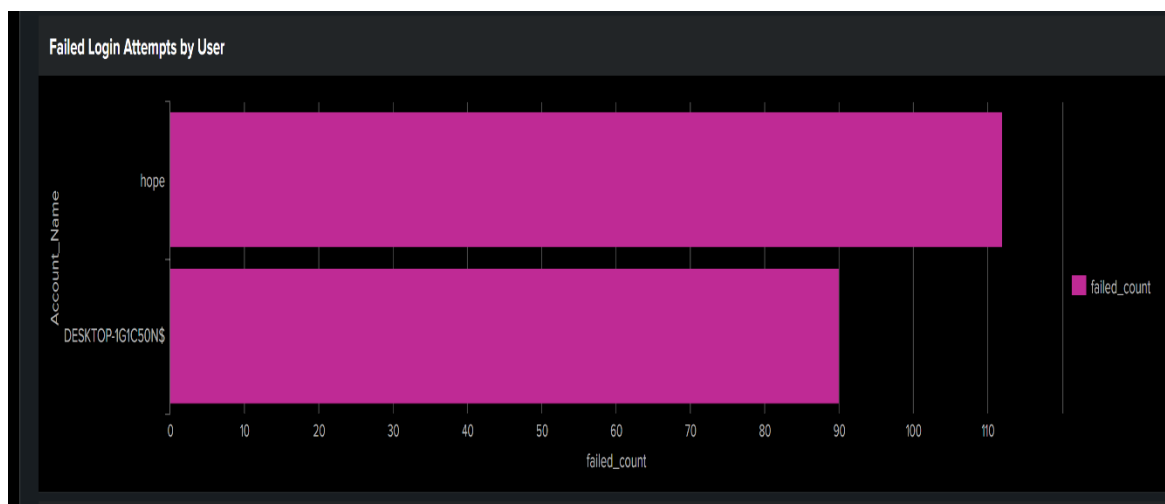
The following actions are recommended:

- Verify with the user whether the failed login attempts were expected.
- Check whether a successful login occurred after the failed attempts.
- Review the source workstation and affected account.
- Reset the password if the activity is suspicious.
- Enable or review account lockout policy.
- Continue monitoring for repeated failed login attempts.
- Escalate to higher-level SOC analyst if suspicious activity continues.

10. Conclusion

The Splunk detection successfully identified multiple failed login attempts using Windows Security Event ID 4625. The alert was validated through Splunk Triggered Alerts and reviewed using Splunk Search & Reporting.

This incident demonstrates basic SOC investigation skills, including alert validation, log analysis, SPL query usage, event correlation, MITRE ATT&CK mapping, and remediation recommendation.



splunk>Enterprise

Search & Reporting

Search

Analytics

Datasets

Reports

Alerts

Dashboards

Modules

New Search

Save As

Create Table View

Close

Time range: Last 24 hours

Q

index=soc_lab sourcetype="WinEventLog:Security" EventCode=4625 earliest=-15m latest=now

| where Account_Name!="-*" AND Account_Name!="ANONYMOUS LOGIN"

| stats count as failed_count by Account_Name Workstation_Name host

| where failed_count >= 3

| sort - failed_count

8 events (19/06/2026 15:34:04.000 to 19/06/2026 15:49:04.000)

No Event Sampling

Job

Smart Mode

Events

Patterns

Statistics (2)

Visualization

Show: 20 Per Page

Format

Preview: On

Account_Name	Workstation_Name	host	failed_count
DESKTOP-1G1C50N\$	DESKTOP-1G1C50N	DESKTOP-1G1C50N	8
hope	DESKTOP-1G1C50N	DESKTOP-1G1C50N	8

splunk>Enterprise

Triggered Alerts

Q Filter

App Search & Reporting (search)

Owner All owners

Severity All severity

Alert name SOC-LAB-001 - Multiple Failed Login Attempts

Showing 1 - 12 of 12 results

Show: 20 per page

1

of 1 pages

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Time	Alert name	App	Type	Severity	Mode	Actions
2026-06-19 22:35:01 India Standard Time	SOC-LAB-001 - Multiple Failed Login Attempts	search	Scheduled	Medium	Digest	View Results Edit Alert Delete
2026-06-19 22:30:01 India Standard Time	SOC-LAB-001 - Multiple Failed Login Attempts	search	Scheduled	Medium	Digest	View Results Edit Alert Delete
2026-06-19 22:25:01 India Standard Time	SOC-LAB-001 - Multiple Failed Login Attempts	search	Scheduled	Medium	Digest	View Results Edit Alert Delete
2026-06-19 17:15:02 India Standard Time	SOC-LAB-001 - Multiple Failed Login Attempts	search	Scheduled	Medium	Digest	View Results Edit Alert Delete
2026-06-19 17:10:01 India Standard Time	SOC-LAB-001 - Multiple Failed Login Attempts	search	Scheduled	Medium	Digest	View Results Edit Alert Delete
2026-06-19 17:05:01 India Standard Time	SOC-LAB-001 - Multiple Failed Login Attempts	search	Scheduled	Medium	Digest	View Results Edit Alert Delete
2026-06-19 16:55:01 India Standard Time	SOC-LAB-001 - Multiple Failed Login Attempts	search	Scheduled	Medium	Digest	View Results Edit Alert Delete
2026-06-19 16:50:02 India Standard Time	SOC-LAB-001 - Multiple Failed Login Attempts	search	Scheduled	Medium	Digest	View Results Edit Alert Delete
2026-06-19 16:45:01 India Standard Time	SOC-LAB-001 - Multiple Failed Login Attempts	search	Scheduled	Medium	Digest	View Results Edit Alert Delete
2026-06-19 16:25:02 India Standard Time	SOC-LAB-001 - Multiple Failed Login Attempts	search	Scheduled	Medium	Digest	View Results Edit Alert Delete
2026-06-19 16:20:01 India Standard Time	SOC-LAB-001 - Multiple Failed Login Attempts	search	Scheduled	Medium	Digest	View Results Edit Alert Delete
2026-06-19 16:15:01 India Standard Time	SOC-LAB-001 - Multiple Failed Login Attempts	search	Scheduled	Medium	Digest	View Results Edit Alert Delete